▷ **Dividing roots**
Dividing one number by another and taking the square root of the result is the same as dividing the square root of the first number by the square root of the second.

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}} \quad \Rightarrow \quad \sqrt{\frac{7}{16}} = \frac{\sqrt{7}}{\sqrt{16}} = \frac{\sqrt{7}}{4}$$

√7 is irrational (2.6457...), so leave as a surd

16 is 4 squared

▷ **Simplifying further**
When dividing square roots, look out for ways to simplify the top as well as the bottom of the fraction.

$$\sqrt{\frac{8}{9}} = \frac{\sqrt{8}}{\sqrt{9}} = \frac{\sqrt{8}}{3} = \frac{2 \times \sqrt{2}}{3}$$

√9 = 3 (3 x 3 = 9)

$$\sqrt{8} = \sqrt{4} \times \sqrt{2} = 2 \times \sqrt{2}$$

8 is 4 × 2

4 is 2 squared

final, simplified form

Surds in fractions

When a surd appears in a fraction, it is helpful to make sure it appears in the numerator (top of the fraction) not the denominator (bottom of the fraction). This is called rationalizing, and is done by multiplying the whole fraction by the surd on the bottom.

▷ **Rationalizing**
The fraction stays the same if the top and bottom are multiplied by the same number.

$$\frac{1}{\sqrt{2}} = \frac{1 \times \sqrt{2}}{\sqrt{2} \times \sqrt{2}} = \frac{\sqrt{2}}{2}$$

multiply top and bottom by the surd √2

the surd √2 is now on top of the fraction

▷ **Simplifying further**
Sometimes rationalizing a fraction gives us another surd that can be simplified further.

$$\frac{12}{\sqrt{15}} = \frac{12 \times \sqrt{15}}{\sqrt{15} \times \sqrt{15}} = \frac{12 \times \sqrt{15}}{15} = \frac{4 \times \sqrt{15}}{5}$$

12 and 15 can both be divided by 3 to simplify further

multiply both top and bottom by √15

multiplying √15 by √15 gives 15